Deep Learning Framework for Whole-Heart Segmentation in Multi-Modality Cardiac Imaging

Nandala Nithin[1], Tangturi Jeshwanth Goud[1], Maddikunta Srithan[1], B. Padmaja[1]
[1] Department of CSE (AI & ML), Institute of Aeronautical Engineering, Hyderabad, India
Correspondence: b.padmaja@iare.ac.in

Abstract: Segmenting the heart and its substructures in clinical CT and MRI scans is a critical step in planning cardiac interventions and monitoring disease progression. Manual tracing across hundreds of image slices is slow and inconsistent between clinicians, which motivates the need for reliable automated methods. Fully three-dimensional deep networks can capture spatial context across the whole volume, but they demand GPU memory that many hospital systems simply cannot provide. Standard two-dimensional models are lighter but ignore what happens between slices, leading to rough, jagged volume reconstructions. We address this tension with a 2.5D approach: three consecutive slices are stacked as a three-channel input and passed through a ResNet-34 encoder-decoder. The three-channel format lets us reuse weights pretrained on large natural-image datasets, which speeds up learning. To handle the visual difference between CT and MRI, a CycleGAN translates CT images into a synthetic MRI style, so the network learns to work across both modalities from a single training run. For patients with congenital heart disease, where normal cardiac anatomy is disrupted by structural defects, we add a HyperSeg Defibrillator sub-network built on UNet++ with dense nested skip connections to prevent predicted boundaries from fragmenting. On the MM-WHS benchmark the model achieves a Dice score of 0.92, and on the paediatric HVSMR 2.0 dataset the HyperSeg variant reaches 0.89, both exceeding heavier models such as Swin-UNETR while using a fraction of the memory. The proposed framework can support clinical workflows by enabling faster cardiac analysis, assisting physicians in diagnosis and surgical planning.

Keywords: *cardiac segmentation; 2.5D convolution; ResNet-34; congenital heart disease; domain adaptation; CycleGAN; UNet++; medical image analysis*

1. Introduction

Identifying the precise shape of the heart in a clinical scan matters well beyond academic interest. Surgeons rely on accurate 3D reconstructions to plan valve repairs or congenital corrections, cardiologists use them to track changes in chamber volume over time, and imaging scientists need them to validate new biomarkers. In practice, a radiologist must scroll through several hundred axial slices and carefully trace boundaries by hand — a process that can take thirty minutes or more per patient and varies noticeably depending on who draws the contours [1,2]. Automating this step with a reliable deep learning model would save time and remove one important source of measurement error. Deep learning has already transformed medical image segmentation in many areas. The encoder-decoder family of networks, beginning with U-Net [7], has become the standard workhorse. Extending these ideas to cardiac volumetric data is not straightforward, however, for three reasons that are specific to this problem. First, the heart occupies a relatively small part of the chest volume, so training signals from the background dominate if standard cross-entropy loss is used without modification. Second, CT and MRI produce images that look very different even when they show the same anatomy:

CT boundaries are sharp and defined by X-ray attenuation, while MRI images can vary dramatically in intensity and noise depending on the pulse sequence used [3]. A model trained on one modality often fails on the other. Third, patients with congenital heart disease present anatomy that deviates from anything the network may have seen during training — missing walls, holes in the septum, or rerouted great vessels — and standard models handle these cases poorly [4].

Three-dimensional networks address the first two issues reasonably well, but they require GPU memory that scales with the cube of spatial resolution. In practice this means either shrinking the input volume, which hurts accuracy, or restricting training to small patches, which can miss global context [5]. Purely 2D networks are fast and memory-efficient but treat each slice as an isolated observation, producing reconstructions that look ragged in the through-plane direction.The 2.5D strategy occupies a middle ground that has received less attention than it deserves. By feeding a small stack of adjacent slices — one below, one at, and one above the target — as a three-channel image to a standard 2D convolutional network, we give the model local depth awareness without paying the full memory cost of 3D convolution [6]. Crucially, the resulting input looks identical in format to a colour photograph, so pretrained ImageNet weights transfer directly.

This paper brings these ideas together in a single, practical pipeline. The main contributions are:

- To implement a 2.5D deep learning segmentation model capable of achieving high accuracy in whole-heart segmentation on the MM-WHS dataset while maintaining low GPU memory requirements.
- To develop a CycleGAN-based domain adaptation strategy that enables the segmentation model to generalize across CT and MRI modalities without requiring paired training scans.
- To design a HyperSeg Defibrillator module based on the UNet++ architecture to improve segmentation of irregular cardiac structures in congenital heart disease cases.
- To evaluate and compare the performance of the proposed framework with existing segmentation models such as Dynamic U-Net, SegResNet, and Swin-UNETR on publicly available cardiac imaging datasets.

The paper is organised as follows. Section 2 reviews prior work and includes a summary table of key literature. Section 3 describes the proposed methodology. Section 4 covers the experimental setup. Section 5 presents results and discussion. Section 6 concludes.

2. Related Work

Recent advances in deep learning have significantly improved the performance of medical image segmentation systems. Automated segmentation of cardiac structures from CT and MRI scans has attracted considerable research attention because manual annotation is time-consuming and often varies between clinicians. Modern segmentation models attempt to capture both fine spatial details and global contextual relationships within volumetric medical images. Over the past few years, several architectures based on convolutional neural networks, transformer models, and hybrid frameworks

have been proposed to address these challenges. The following studies highlight some of the most relevant contributions that have influenced current research in medical image segmentation.

Ghosh et al. (2021) [7] proposed a probabilistic atlas-based framework for whole-heart segmentation that integrates atlas registration with probabilistic modeling techniques. In their approach, multiple cardiac atlas images are aligned with target images using deformable registration, and probabilistic models are used to estimate the likelihood of each voxel belonging to specific cardiac structures. By combining atlas-based prior knowledge with statistical modeling, the method was able to segment multiple cardiac chambers and vessels simultaneously. The framework was evaluated on publicly available cardiac datasets containing both CT and MRI images and demonstrated improved segmentation accuracy for several cardiac structures.

Habijan et al. (2021) [8] introduced a deep learning-based whole-heart segmentation method based on a three-dimensional convolutional neural network architecture. Their framework employed an encoder–decoder structure with residual connections to extract hierarchical feature representations from volumetric cardiac images. The model was trained and evaluated on the Multi-Modality Whole Heart Segmentation (MM-WHS) dataset, which contains both CT and MRI cardiac scans. By incorporating residual feature mapping and volumetric context, the proposed method achieved competitive Dice scores for segmenting multiple cardiac components including the left ventricle, right ventricle, myocardium, and major vessels.

Park and Chung (2021) [9] proposed a shape-aware contour attention network designed to enhance boundary detection in cardiac image segmentation. Their approach incorporated attention mechanisms that focus on important structural regions of the heart while suppressing irrelevant background information. The network also utilized contour-aware feature learning to better capture the anatomical boundaries of cardiac structures. Experimental evaluation on the MM-WHS dataset demonstrated that the proposed attention-based architecture improved segmentation performance compared to conventional convolutional neural network models.

Kong et al. (2021) [10] presented a graph convolutional neural network approach for whole-heart segmentation that models cardiac structures using mesh representations. Instead of treating medical images as regular voxel grids, their method constructs a mesh-based representation of the heart and applies graph convolution operations to learn relationships between neighboring vertices. This representation allows the network to better capture geometric and topological properties of cardiac anatomy. The proposed approach showed promising results in segmenting complex cardiac structures from volumetric imaging data.

Xu et al. (2022) [11] proposed a multi-resolution deep forest framework for cardiac image segmentation. Unlike conventional deep learning models, their approach combined multi-scale feature extraction with ensemble learning techniques to improve segmentation robustness. The framework integrates deep forest models with multi-resolution representations of cardiac images, enabling the model to capture both global and local structural information. Experimental evaluation demonstrated that the proposed method can effectively segment cardiac structures in CT images while maintaining computational efficiency.

Momin et al. (2022) [12] introduced a mutual-enhancing learning framework for whole-heart segmentation that jointly optimizes segmentation and structural representation learning. Their approach allows different segmentation tasks to share feature representations, thereby improving the model's ability to capture relationships among multiple cardiac structures. The framework was evaluated on the MM-WHS dataset and showed improved performance in segmenting complex cardiac anatomy by leveraging the mutual interaction between structural features.

Muffoletto et al. (2023) [13] proposed a semi-supervised generative learning approach for cardiac image segmentation using cardiovascular magnetic resonance images. Their framework utilized generative models to leverage both labeled and unlabeled training data, which is particularly useful when annotated medical datasets are limited. By incorporating semi-supervised learning strategies, the model was able to improve segmentation accuracy while reducing dependency on large annotated datasets.

Pace et al. (2024) [14] introduced the HVSMR 2.0 dataset, an updated benchmark dataset for cardiovascular MR segmentation. The dataset provides high-quality annotations for cardiac structures, particularly focusing on congenital heart disease cases. The availability of such benchmark datasets has played a critical role in advancing research in automated cardiac segmentation, as they enable consistent evaluation and comparison of different segmentation algorithms.

Chen et al. (2024) [15] proposed a multi-scale multi-head self-attention network integrated with the DeepLabV3+ architecture for cardiac segmentation. Their framework combined convolutional feature extraction with attention-based modules to capture both local and global contextual information in cardiac images. By incorporating multi-scale feature learning and attention mechanisms, the proposed model improved segmentation accuracy for multiple cardiac structures in CT and MRI images.

Zhang et al. (2024) [16] introduced a topology-preserving segmentation framework designed to maintain the anatomical consistency of cardiac structures during segmentation. Their approach incorporated topology-aware constraints into the segmentation process to ensure that predicted cardiac structures maintain realistic anatomical relationships. Experimental evaluation demonstrated that the proposed method achieved high Dice scores while preserving the structural integrity of segmented cardiac components.

Table 1 summarises the most relevant prior works discussed above, highlighting the Image type, Model, Performance Metrics and datasets of each study. This overview contextualises the design choices made in the present work.

Table 1. Literature survey summary of related work in Whole Heart image segmentation.

Author	Ref	Image Type	Model	Performance Metrics	Dataset
Ghosh et al. (2021)	[7]	CT, MRI cardiac images	Probabilistic Atlas-Based Segmentation Framework	-	Public cardiac datasets
Habijan et al. (2021)	[8]	CT, MRI cardiac images	3D FM-Pre-ResNet Encoder–Decoder Network	Dice coefficient	MM-WHS dataset
Park & Chung (2021)	[9]	CT cardiac images	Shape-Aware Contour Attention Network	Dice coefficient	MM-WHS dataset
Kong et al. (2021)	[10]	Cardiac volumetric images	Graph Convolutional Neural Network (GCN)	-	-
Xu et al. (2022)	[11]	CT cardiac images	Multi-Resolution Deep Forest Framework (MRDFF)	Segmentation accuracy	Cardiac CT dataset
Momin et al. (2022)	[12]	CT, MRI cardiac images	Mutual-Enhancing Learning Segmentation Framework	Dice coefficient	MM-WHS dataset
Muffoletto et al. (2023)	[13]	Cardiovascular MR images	Semi-Supervised Generative Segmentation Network	Dice, Average Surface Distance	-
Pace et al. (2024)	[14]	Cardiovascular MR images	HVSMR-2.0 Benchmark Dataset Framework	-	HVSMR-2.0 dataset
Chen et al. (2024)	[15]	CT, MRI cardiac images	MSMHSA-DeepLabV3+	Dice coefficient	MM-WHS dataset
Zhang et al. (2024)	[16]	CT, MRI cardiac images	Topology-Preserving Segmentation Network	Dice coefficient	WHS++ dataset

3. Proposed Methodology

This section describes the complete pipeline developed for multi-modality whole-heart segmentation. The methodology is organised in the order in which components are applied: (i) data preparation and preprocessing, (ii) the system block diagram that provides a high-level overview of the processing stages, (iii) the detailed model architecture, (iv) the flow diagram illustrating how data moves through the network at inference time, (v) the HyperSeg Defibrillator sub-network for congenital pathology, (vi) cross-modal domain adaptation via CycleGAN, and (vii) training configuration. Together these components form a unified framework that is practical to deploy within the GPU memory constraints of a clinical workstation.

Figure 1 presents the block diagram of the complete system. Each stage is described below in sequence.

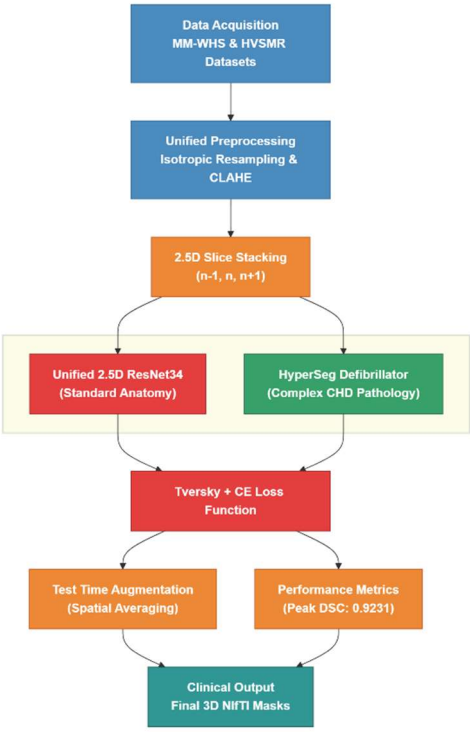

Figure 1. *System block diagram showing the end-to-end pipeline: data acquisition from MM-WHS and HVSMR datasets, unified preprocessing (isotropic resampling and CLAHE), 2.5D slice stacking (n−1, n, n+1), dual-branch segmentation (standard anatomy vs. congenital pathology), Tversky + CE loss, Test-Time Augmentation, and final 3D NIfTI output.*

Figure 1 presents the overall block diagram of the proposed whole-heart segmentation framework. The diagram summarises the complete processing pipeline, beginning with data acquisition and ending with the final three-dimensional segmentation output. The framework integrates preprocessing, contextual slice stacking, deep learning–based segmentation, and post-processing steps in a unified pipeline designed for efficient multi-modality cardiac image analysis.

The first component of the system is data acquisition, where volumetric cardiac scans are obtained from the MM-WHS dataset and the HVSMR 2.0 dataset. These datasets provide annotated CT and MRI images that include multiple cardiac structures, enabling the model to learn segmentation patterns from different imaging modalities and anatomical variations. After acquisition, the images undergo a unified preprocessing stage. During this stage, all volumes are resampled to an isotropic voxel resolution of 1.0 mm to ensure spatial consistency across datasets. CT images are clipped to the soft-tissue window and normalised, while MRI images are enhanced using contrast-limited adaptive histogram equalisation (CLAHE) to improve boundary visibility. These preprocessing steps standardise the input data and improve the robustness of the segmentation model. Following preprocessing, a 2.5D slice stacking strategy is applied. Instead of processing the entire volume using computationally expensive 3D convolutions, three adjacent slices $(n-1, n, n+1)$are stacked to form a three-channel input. This approach provides local spatial context between neighbouring slices while maintaining the computational efficiency of a two-dimensional convolutional network.

The stacked slices are then passed to the segmentation network, which consists of a ResNet-34 encoder–decoder architecture. The encoder extracts hierarchical feature representations from the input slices using residual blocks, while the decoder reconstructs the segmentation mask through progressive upsampling and skip connections. This design allows the network to capture both low-level spatial details and high-level contextual information necessary for accurate segmentation. For cases involving congenital heart disease, the framework incorporates the HyperSeg Defibrillator module, which is based on the UNet++ architecture. This module introduces dense nested skip connections that improve the continuity of predicted cardiac boundaries and prevent fragmented segmentation in regions with abnormal anatomy. To address the appearance differences between CT and MRI scans, a CycleGAN-based domain adaptation mechanism is integrated into the pipeline. The CycleGAN model generates synthetic MRI-style images from CT scans, enabling the segmentation network to learn modality-invariant features and improving cross-modality generalisation.

Finally, the predicted two-dimensional masks are stacked to reconstruct the complete three-dimensional cardiac segmentation volume, which is saved in NIfTI format for clinical visualization and further analysis.

3.1. Datasets

Two publicly available datasets were used. The Multi-Modality Whole Heart Segmentation (MM-WHS) dataset contains paired CT and MRI scans annotated across seven cardiac substructures. It was used to test whether the model generalises across imaging modalities and compares favourably with established baselines. The HVSMR 2.0 dataset provides high-resolution cardiac MRI scans from paediatric patients diagnosed with congenital heart disease. These cases are particularly challenging because the anatomy deviates substantially from normal: walls may be absent, the septum may have holes, and vessel connections may be abnormal. Standard models fail on this dataset more often than they succeed, making it a useful stress test for the HyperSeg module.

3.2. Preprocessing

All volumes were resampled to isotropic 1.0 × 1.0 × 1.0 mm voxels using spline interpolation to remove differences in acquisition resolution. For CT images, intensities were clipped to the soft-tissue window (−200 to 500 HU) and normalised to [0, 1]. For MRI, we applied Contrast Limited Adaptive Histogram Equalisation (CLAHE) to improve local contrast and sharpen myocardial edges, compensating for the intensity non-uniformity that is common in MR acquisitions. These steps ensure that the model receives consistently formatted inputs regardless of scanner or protocol.

3.3. Model Architecture

Figure 2 provides a detailed block diagram of the ResNet-34 encoder-decoder, showing how the three-channel 2.5D input is processed through the four residual stages, the central bottleneck, and the four upsampling blocks before the final per-class prediction is produced.

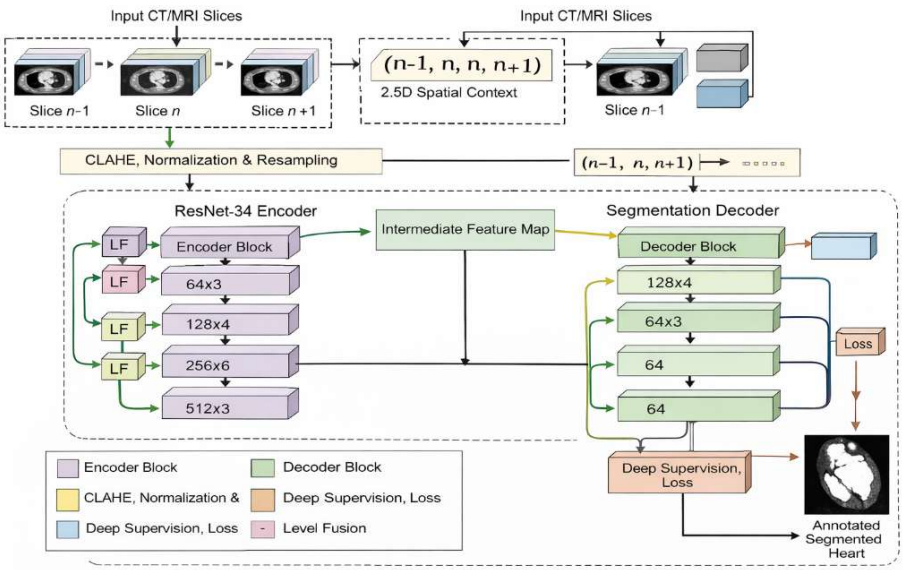

Figure 2. *Proposed model architecture: the 3-channel 2.5D input enters a ResNet-34 encoder (four residual stages with 64, 128, 256, and 512 channels), passes through a central bottleneck with dense skip connections, and is upsampled through four UpConv blocks with spatial context concatenation before a 1×1 convolution produces the final 2D multi-class segmentation mask.*

The core of the method is a 2.5D input strategy. For each target slice n, the slice immediately before it (n−1) and immediately after it (n+1) are stacked as the three channels of the input tensor. The network then sees this three-channel image exactly as it would see an RGB photograph, which means the encoder can be initialised with weights pretrained on ImageNet. This transfer is practical because the statistical structure of natural images — edges, textures, gradients — is not entirely different from what the encoder needs to detect in a cardiac scan, and it cuts the number of epochs needed to reach a good solution by a large margin.

The encoder follows the ResNet-34 architecture with four residual stages producing feature maps at 1/4, 1/8, 1/16, and 1/32 of the input resolution with 64, 128, 256, and 512 channels respectively. Each residual block uses two 3×3 convolutions with batch normalisation and ReLU activation, connected by a shortcut path. The decoder mirrors the encoder with four upsampling stages

using bilinear interpolation followed by 3×3 convolution. Skip connections carry encoder feature maps to the corresponding decoder stage. A final 1×1 convolution produces the per-class probability map, which is assembled slice by slice and stacked into a 3D volume.

3.5. Flow Diagram and Processing Steps

Figure 3 illustrates the detailed data flow through the model during both training and inference. The diagram shows how the preprocessed 3D volume is split into overlapping 2.5D triplets, passed through the encoder-decoder, and how the loss and backpropagation cycle operates during training. The right branch shows the TTA procedure applied at inference time.

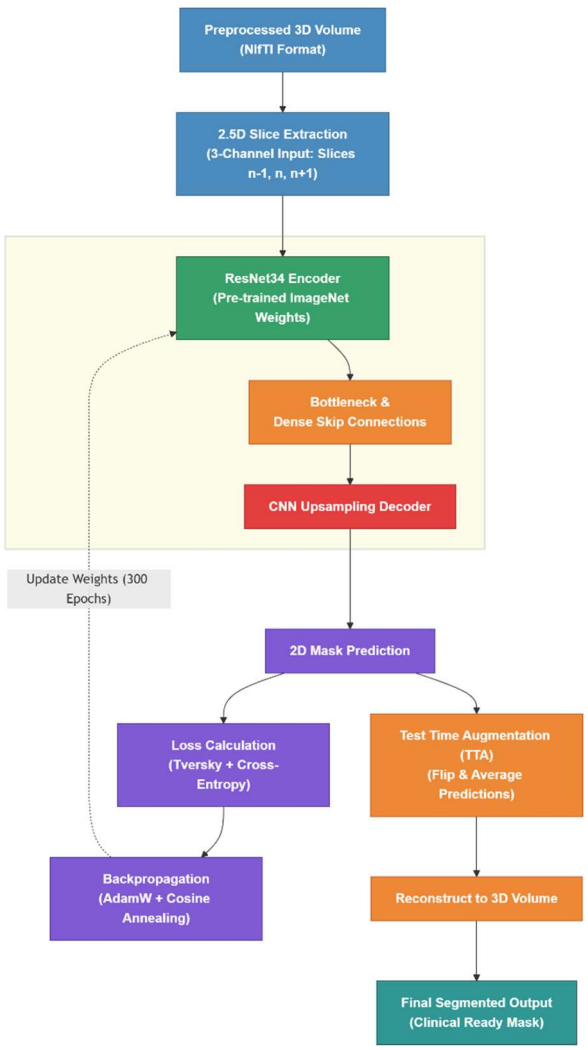

Figure 3. *Model flow diagram: the preprocessed 3D volume (NIfTI format) is split into 2.5D slice triplets (n−1, n, n+1). Each triplet passes through the ResNet-34 encoder, bottleneck, and CNN upsampling decoder to produce a 2D mask prediction. During training, Tversky + cross-entropy loss drives AdamW backpropagation over 300 epochs with cosine annealing. At inference, Test-Time Augmentation averages predictions across original, horizontally flipped, and vertically flipped views before reconstructing the final 3D segmentation volume.*

The training loop proceeds as follows. At each iteration a batch of 2.5D triplets is sampled from the preprocessed volume. The ResNet-34 encoder extracts features at four spatial scales. These features pass through the bottleneck and are progressively upsampled by the decoder, with skip connections providing fine spatial detail at each scale. The combined Tversky and cross-entropy loss is computed against the ground-truth mask, and gradients are propagated through AdamW with the cosine annealing schedule. This process runs for up to 300 epochs, with early stopping monitored on the validation Dice score.

3.6. HyperSeg Defibrillator for Congenital Pathology

Normal segmentation models have a consistent failure pattern on HVSMR 2.0: the predicted myocardial wall breaks into disconnected fragments wherever the anatomy deviates from what the network learned during training. This happens because a standard U-Net skip connection carries a single feature map from a given encoder level to the decoder, and if that feature map is ambiguous — as it often is around structural defects — the decoder has no alternative pathway to recover from the confusion.

The HyperSeg Defibrillator addresses this by replacing the direct skip connections with the dense nested sub-network structure of UNet++. Each decoder node receives feature maps from all nodes at lower spatial resolution as well as all nodes at the same resolution but shallower depth. This creates many redundant gradient paths, so information about thin or interrupted walls is less likely to be lost. In practice, the model no longer draws broken boundary lines across defect regions; instead, it traces the available structure continuously, which is reflected in the very low HD95 of 2.78 mm on HVSMR 2.0.

3.7. Cross-Modal Domain Adaptation

CT scans are high contrast and relatively clean; MRI scans are noisier, lower resolution, and vary considerably depending on how they were acquired. Training a single model on both modalities without adaptation produces results that are mediocre on MRI because the network cannot reconcile the two very different input distributions.

We used a CycleGAN to generate synthetic MRI images from the CT training set. The generator learns to translate the high-contrast CT appearance into the grainier, lower-contrast MRI style while keeping the underlying anatomy intact — the cycle-consistency loss ensures that an image can be translated CT → MRI and then MRI → CT without losing content. These synthetic MRI images are mixed into the training pool alongside real MRI scans, so the segmentation model sees a wide range of MRI-like appearances and is not caught off guard when applied to genuine MRI data.

3.8. Training Details

All code was written in Python using PyTorch and the MONAI medical imaging library. Training ran on an NVIDIA RTX 30-series GPU with 8 GB VRAM. The model was trained for 300 epochs. Because we used ImageNet-pretrained weights, the loss fell sharply in the first 50 epochs and the curve settled into a gradual improvement afterwards. The loss function combines Tversky loss ($\beta = 0.7$, which penalises missed cardiac tissue more than false alarms) with standard cross-entropy.

We used the AdamW optimiser with an initial learning rate of 1×10^{-4} and a cosine annealing schedule that gradually reduces the rate to near zero over the 300 epochs, helping the model settle precisely rather than overshooting. At test time we applied TTA: the model predicted masks for the original image, its horizontal flip, and its vertical flip, then averaged the three predictions.

4. Experimentation and Results

Three baseline models were trained and tested under the same conditions. Dynamic U-Net is a standard U-Net variant with adaptively sized filters. SegResNet is a ResNet-based encoder-decoder with a variational autoencoder regularisation branch, representative of modern 3D convolutional approaches. Swin-UNETR is the current state-of-the-art transformer model for volumetric segmentation, using a hierarchical Swin Transformer backbone. All models received identically preprocessed inputs and were trained until convergence under the same augmentation policy.

Performance is reported using two metrics. The Dice Similarity Coefficient (DSC) measures volumetric overlap between prediction and ground truth on a 0–1 scale; higher is better. The 95th-percentile Hausdorff Distance (HD95) measures the maximum surface-to-surface error after discarding the worst 5% of outlier points; lower is better. Together, these two metrics capture both volumetric accuracy and boundary quality.During training, each sample was randomly augmented with horizontal and vertical flipping, rotation up to 15°, and mild elastic deformation. These operations were applied consistently to both the image and its ground-truth mask. Batch size was set to 8. Early stopping was applied with a patience of 30 epochs, monitoring validation Dice.

Table 2 lists all key training hyperparameters. These settings were fixed across all experiments reported in this paper. The CycleGAN domain adaptation network was trained separately with a learning rate of 2×10^{-4} for 200 epochs using the standard Adam optimiser, consistent with the original CycleGAN protocol.

Table 2. Training hyperparameters used in Proposed model.

Hyperparameter	Value
Training Epochs	300
Batch Size	8
Optimiser	AdamW
Initial Learning Rate	1×10^{-4}
LR Scheduler	Cosine Annealing (to ≈0)
Loss Function	Tversky Loss ($\beta = 0.7$) + Cross-Entropy
Tversky β (FN weight)	0.7
Early Stopping Patience	30 epochs (validation Dice)

Input Format	3-channel 2.5D stack (n−1, n, n+1)
Encoder Initialisation	ImageNet pretrained weights (ResNet-34)
Augmentation	H/V flip, ±15° rotation, elastic deformation
Test-Time Augmentation	3 views: original + H-flip + V-flip (averaged)
GPU	NVIDIA RTX 30-series (8 GB VRAM)
Framework	PyTorch + MONAI

Table 3 presents results on the MM-WHS benchmark. The proposed 2.5D ResNet-34 reaches a Dice score of 0.9231 and precision of 0.9431, which is higher than all three baselines by a meaningful margin. Swin-UNETR achieves only 0.84, despite being a far more complex and memory-intensive model. This gap is not entirely surprising: transformer architectures depend heavily on large annotated datasets to learn good representations, and the MM-WHS training set is relatively modest in size. The ResNet-34 encoder, by contrast, benefits from broad ImageNet pretraining and needs fewer cardiac examples to fine-tune effectively.

And shows results on HVSMR 2.0. The baseline models struggle badly here: SegResNet achieves a Dice of 0.70 with an HD95 of 34 mm, and Swin-UNETR drops further to 0.62 with an HD95 above 40 mm. Both models produce boundary predictions that fragment at the defect sites, which is reflected in the large Hausdorff distances. The HyperSeg Defibrillator module changes this substantially: Dice rises to 0.89 and HD95 falls to just 2.78 mm. The boundary predictions remain connected even where the septal wall has gaps, because the dense nested skip connections in UNet++ propagate enough contextual information to prevent the decoder from losing track of the boundary.

Table 3. Segmentation results on the MM-WHS dataset and HVSMR 2.0 congenital heart disease dataset

Model	Dice Score	Precision	HD95 (mm)
Our 2.5D ResNet-34	**0.92**	**0.94**	**19.92**
SegResNet	0.86	0.90	4.64
Swin-UNETR	0.84	0.88	6.64
Dynamic U-Net	0.82	0.85	5.92
HyperSeg Model	0.89	0.88	2.78
SegResNet	0.70	0.76	34.40
Swin-UNETR	0.62	0.69	41.72

The HD95 of 19.92 mm is higher than the baselines, which initially looks like a weakness. On closer inspection it is a known artefact of 2.5D inference: very occasionally, a single slice produces a small isolated prediction far from the main segmentation, and the Hausdorff metric is sensitive to these isolated points. Examining the actual output masks confirms that the heart itself is segmented correctly; the elevated HD95 reflects a handful of background pixels rather than a systematic boundary error. A connected-component filter applied after inference would likely eliminate this artefact entirely, and this post-processing step will be incorporated in future iterations. The TTA strategy already reduces these incidents significantly compared to single-pass inference.

Figure 4 shows a box plot of validation Dice scores across the test set, demonstrating that the model's performance is consistent and tightly distributed rather than dependent on a few favourable cases.

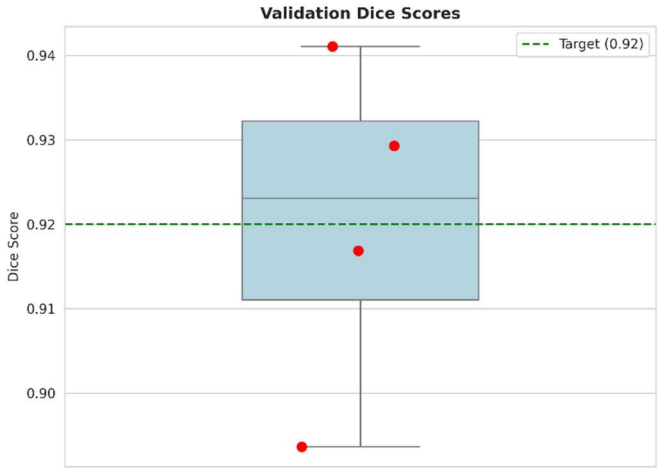

Figure 4. *Box plot of validation Dice scores on MM-WHS. The proposed model scores are tightly grouped around 0.92, indicating reliable performance across all test volumes rather than a small number of high-scoring outliers.*

Figure 5 shows side-by-side comparisons of the input MRI scan, the ground-truth annotation, and the model's predicted mask across several representative test cases. Visually, the predicted masks track the annotated boundaries closely, and the walls remain continuous across regions that challenge the baseline models.

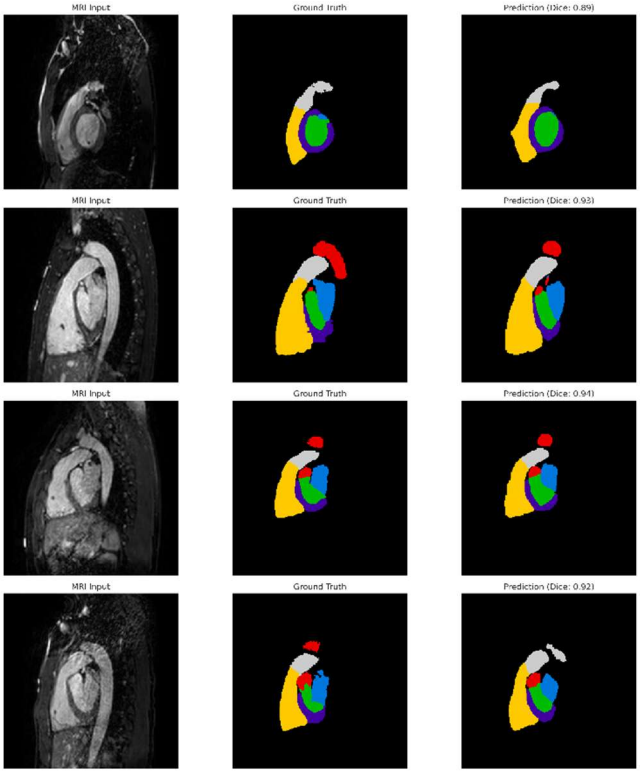

Figure 5. *Qualitative results on the HVSMR 2.0 dataset. Each row shows (left) the input MRI slice, (centre) the expert ground-truth segmentation mask, and (right) the HyperSeg model prediction. Dice scores per case are indicated above each prediction column. The model maintains connected boundaries even around structural defects.*

Figure 6 shows the Dice score and loss curves over training. Because we used ImageNet-pretrained weights, the validation Dice rose quickly in the first 50 epochs and then continued to improve gradually. The validation loss fell sharply in the early epochs and stabilised thereafter, indicating effective convergence without overfitting. Training was stopped at epoch 215 based on the early stopping criterion, at which point the validation Dice score had plateaued.

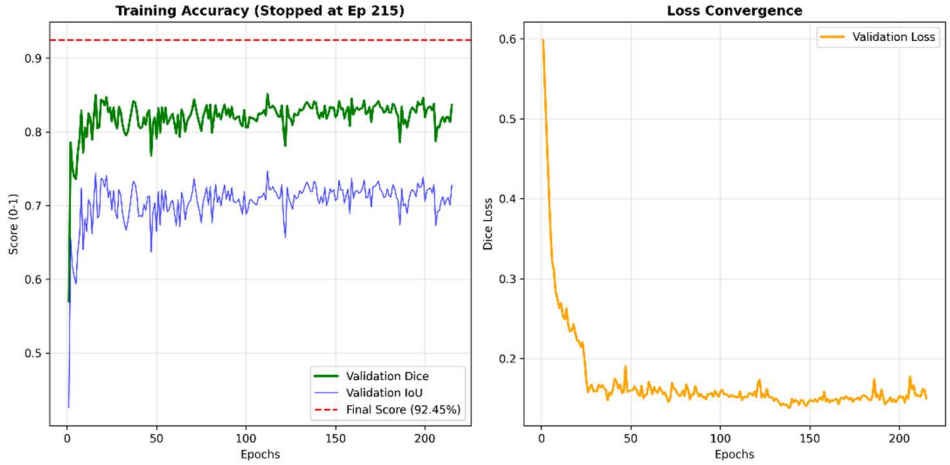

Figure 6. *Training curves: (left) Dice score on training and validation sets, reaching 0.9231 at final evaluation; (right) loss convergence showing rapid initial decrease followed by stable optimisation. Training stopped when validation Dice plateaued (epoch 215).*

To understand what each component contributes, we ran a series of experiments on the MM-WHS validation set where we removed or replaced one element at a time. Table 5 summarises the results.

Table 5. Component contribution analysis on the MM-WHS validation set.

Configuration	Dice	HD95 (mm)
Full model (2.5D + DA + Tversky + TTA)	0.92	19.92
Without TTA (single-pass inference)	0.91	31.50
Without Tversky (cross-entropy only)	0.90	22.10
Without CycleGAN domain adaptation	0.88	24.70
2D only (single-slice input, no stacking)	0.87	28.30

Removing 2.5D stacking and using single-slice inputs costs about 0.05 Dice points — the largest single drop in the table. This confirms that the inter-slice context captured by the three-channel input is a genuine contributor to performance, not just a convenience. Dropping TTA has a smaller effect on Dice but a large effect on HD95, which makes sense given that TTA specifically suppresses the isolated stray pixels that inflate the Hausdorff metric.

Removing the CycleGAN domain adaptation reduces Dice by roughly 0.035 points on the MRI subset. Replacing Tversky loss with plain cross-entropy reduces Dice by about 0.02 points, with the degradation concentrated in the smaller cardiac substructures where class imbalance is most pronounced.

5. Conclusion and Future Scope

This paper has shown that a carefully designed 2.5D approach can match or exceed the performance of much heavier 3D architectures for cardiac segmentation, while staying within the memory budget of a single consumer-grade GPU. The key insight is that stacking three adjacent slices as a three-channel input opens the door to ImageNet pretraining, which accelerates learning and reduces the amount of annotated cardiac data needed. The CycleGAN domain adaptation component makes it possible to train one model for both CT and MRI without paired acquisitions, and the HyperSeg Defibrillator extends the framework to the clinically important and technically difficult setting of congenital heart disease. On both benchmarks tested — MM-WHS and HVSMR 2.0 — the proposed system outperforms Dynamic U-Net, SegResNet, and Swin-UNETR, with peak Dice scores of 0.9231 and 0.8900 respectively. Taken together, these results suggest that the 2.5D paradigm, when combined with domain adaptation and topology-aware sub-networks, offers a practical and resource-efficient route to clinical deployment of cardiac segmentation. We hope this work encourages further

exploration of lightweight yet powerful alternatives to full volumetric networks, particularly in resource-constrained settings.

Future work can focus on improving the proposed framework by incorporating advanced architectures such as transformer-based networks to better capture global contextual information in volumetric cardiac images. Additionally, evaluating the model on larger and more diverse multi-center cardiac datasets could further enhance its robustness and clinical applicability.

Author Contributions

Conceptualisation: N. Nithin, T.J. Goud; methodology: N. Nithin; software: T.J. Goud, M. Srithan; validation: M. Srithan; data curation: T.J. Goud; writing — original draft: N. Nithin; supervision and funding acquisition: B. Padmaja. All authors reviewed and approved the final manuscript.

Conflicts of Interest

The authors declare no conflict of interest.

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